Identical Information Spectra, Different Causal Feasibility

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Abstract

We exhibit two binary input–output systems with identical conditional information spectra and identical optimal centralized seed entropy, but different feasibility under spatial locality. One admits an entropy-optimal local realization; the other admits no local realization above the CHSH threshold. Its exact distance from the local set, in worst-setting total variation, is computed. This gives an explicit obstruction to extending a spectrum-only causal simulation theorem to distributed decoders. The argument uses established Bell–CHSH theory and does not assert historical novelty or Lean verification.

1 Models and costs

Inputs x, y and outputs a, b belong to $\{0, 1\}$. A behavior is a family of conditional laws $W(a, b \mid x, y)$.

A centralized realization consists of a discrete random seed S, independent of the inputs, and a deterministic decoder

$$(A, B) = F(S, x, y).$$

Its optimal Shannon seed cost is

$$C_{\mathrm{cen}}(W) = \inf H(S),$$

where the infimum ranges over exact centralized realizations.

A local realization additionally requires

$$A = f(S, x), \qquad B = g(S, y).$$

Its optimal cost is denoted by $C_{\mathrm{loc}}(W)$. The infimum of an empty feasible set is $+\infty$. All randomness is included in S; the local decoders do not communicate. Entropies are measured in bits.

Let $\mathcal{L}$ be the set of local behaviors, allowing arbitrary shared randomness independent of the inputs. In this finite scenario, every such behavior is a mixture of finitely many deterministic local response tables.

For $v \in [0, 1]$, define

$$P_v(a, b \mid x, y) = \frac{1 + v(-1)^{a \oplus b}}{4}, \tag{1}$$

$$Q_v(a, b \mid x, y) = \frac{1 + v(-1)^{a \oplus b \oplus xy}}{4}. \tag{2}$$

Both have uniform individual output marginals and satisfy the no-signaling conditions.

2 Spectral equivalence and locality obstruction

Theorem 1. *For every $v \in [0,1]$ and every input pair (x,y), the laws $P_v(\cdot \mid x,y)$ and $Q_v(\cdot \mid x,y)$ have the same multiset of atom masses:*

$$\left\{\frac{1+v}{4}, \frac{1+v}{4}, \frac{1-v}{4}, \frac{1-v}{4}\right\}.$$

Consequently their information distributions, and all their conditional Rényi entropies, coincide.

Writing

$$h(v) = 1 + h_2\left(\frac{1-v}{2}\right),$$

their centralized costs satisfy

$$C_{\mathrm{cen}}(P_v) = C_{\mathrm{cen}}(Q_v) = h(v).$$

Moreover,

$$C_{\mathrm{loc}}(P_v) = h(v),$$

whereas for $v > 1/2$,

$$C_{\mathrm{loc}}(Q_v) = +\infty.$$

In the last assertion no local realization exists even with unrestricted, possibly non-atomic, shared randomness.

Proof. The multiset assertion follows directly from the definitions. Zero-mass atoms at $v = 1$ are omitted when defining information.

Take independent random bits

$$R \sim \mathrm{Bernoulli}(1/2), \qquad E \sim \mathrm{Bernoulli}\left(\frac{1-v}{2}\right).$$

The local functions

$$A = R, \qquad B = R \oplus E$$

realize P_v, using seed entropy $h(v)$. The centralized functions

$$A = R, \qquad B = R \oplus E \oplus xy$$

realize Q_v with the same seed entropy.

For either behavior, the joint output law at any fixed input pair has entropy $h(v)$. It is a deterministic function of the seed, so data processing gives $H(S) \geq h(v)$. This proves the asserted finite optimal costs.

For a behavior W, put

$$E_{xy}(W) = \sum_{a,b} (-1)^{a \oplus b} W(a,b \mid x,y)$$

and

$$\mathcal{B}(W) = E_{00}(W) + E_{01}(W) + E_{10}(W) - E_{11}(W).$$

For a deterministic local table, write $u_x = (-1)^{f(x)}$ and $w_y = (-1)^{g(y)}$. Its CHSH expression is

$$u_0(w_0 + w_1) + u_1(w_0 - w_1) \in \{-2, 2\}.$$

Averaging over any shared seed yields $\mathcal{B}(W) \leq 2$ for every local behavior. But

$$E_{xy}(Q_v) = v(-1)^{xy}, \qquad \mathcal{B}(Q_v) = 4v.$$

Thus Q_v is not local when $v > 1/2$. □

3 The exact approximation threshold

Use the convention

$$\mathrm{TV}(\mu,\nu)=\frac{1}{2}\sum_{z}|\mu(z)-\nu(z)|.$$

Define the distance from a behavior to the local set by

$$d_{\mathrm{loc}}(W)=\inf_{L\in\mathcal{L}}\max_{x,y}\mathrm{TV}\big(W(\cdot\mid x,y),L(\cdot\mid x,y)\big).$$

Theorem 2. *For every $v\in[0,1]$,*

$$\boxed{d_{\mathrm{loc}}(Q_v)=\frac{(v-\frac{1}{2})_+}{2}.}$$

Proof. If a local behavior L is within ε at each input pair, then, since the parity observable is bounded by one,

$$|E_{xy}(Q_v)-E_{xy}(L)|\le 2\varepsilon.$$

Therefore

$$4v=\mathcal{B}(Q_v)\le\mathcal{B}(L)+8\varepsilon\le 2+8\varepsilon.$$

For $v>1/2$ this gives $\varepsilon\ge(v-\frac{1}{2})/2$.

We next construct $Q_{1/2}$ locally. The target parity table xy has odd parity when XORed over the four input pairs. Change exactly one of its four entries. The resulting table t_{xy} has even parity and therefore admits a decomposition

$$t_{xy}=a_x\oplus b_y.$$

Indeed, choose

$$a_0=0,\quad b_0=t_{00},\quad b_1=t_{01},\quad a_1=t_{10}\oplus t_{00};$$

the final entry is correct precisely by even parity.

There are four choices of the changed entry. For each choice, use the two local tables obtained by simultaneously flipping both parties' output bits. Mix these eight deterministic strategies uniformly. At any input pair the target parity holds with probability $3/4$, and the simultaneous flip makes both marginals uniform. The resulting behavior is exactly $Q_{1/2}$.

For $v\ge 1/2$, direct calculation gives

$$\mathrm{TV}\big(Q_v(\cdot\mid x,y),Q_{1/2}(\cdot\mid x,y)\big)=\frac{v-\frac{1}{2}}{2}$$

at every input pair, proving the matching upper bound.

Finally, Q_0 is local: use two independent fair output bits. For $0\le v\le 1/2$,

$$Q_v=2v\,Q_{1/2}+(1-2v)\,Q_0,$$

so Q_v is local and its distance is zero. □

Corollary 3. *At the quantum CHSH value $v=1/\sqrt{2}$,*

$$d_{\mathrm{loc}}(Q_{1/\sqrt{2}})=\frac{\sqrt{2}-1}{4}\approx 0.103553.$$

Thus no amount of shared randomness allows a noncommunicating local simulator to approximate this behavior below that worst-setting total-variation error.

4 Consequence for causal Shannon theory

Corollary 4 (Spectral data do not determine local feasibility)**.** *The collection of conditional information distributions indexed by input pairs does not determine whether a behavior admits a local shared-seed realization. It also does not determine its distance to the local set.*

Proof. Choose $v > 1/2$. The behaviors P_v and Q_v have identical such spectral data. The first is local, so its distance is zero. The second has strictly positive distance by Theorem 2. □

In a centralized controlled-tree model, the pair (x, y) can be regarded as one action and (a, b) as one response. Both behaviors then have the same one-step envelope mean $h(v)$ and attain it exactly.

Spatial locality imposes an additional restriction: each decoder sees only its own input. This restriction concerns how output labels are related across different input pairs. Sorting probabilities into information spectra discards those relations.

Consequently, extending a centralized causal Shannon theorem to distributed simulation requires information about the permitted input dependence of each decoder, in addition to the probability spectra.

The example is not a conflict with no-signaling. No-signaling constrains observable marginal distributions; it does not imply a representation by local deterministic functions of an input-independent shared seed.

5 Scope and attribution

The CHSH obstruction is classical established theory. The inability of entropic data alone to distinguish certain local and nonlocal behaviors also has explicit precedents. This note packages those facts as a precise limitation of spectrum-based seed-cost descriptions.

It does not claim a new Bell theorem, an explanation of the physical mechanism of entanglement, or a Lean-verified result. The family Q_v for $v > 1/\sqrt{2}$ is used only as a mathematical no-signaling family; those parameters violate the quantum CHSH bound.

References

[1] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed Experiment to Test Local Hidden-Variable Theories," Physical Review Letters **23**, 880–884 (1969). `https://doi.org/10.1103/PhysRevLett.23.880`

[2] "Entropic inequalities as a necessary and sufficient condition to noncontextuality and locality," Physical Review A **87**, 022102 (2013). `https://doi.org/10.1103/PhysRevA.87.022102`

[3] J. Mikulík, *One Seed for Every Adaptive Policy.* `https://github.com/jendamikulik/finite-Shannon-theorem`